

AI ENGINEER STUDY PLAN 6 MONTHS

MONTH 1: Python, Git & Engineering Basics

Week	Focus	What You Learn (Detailed)	Build / Output
Week 1	Python Fundamentals	Python syntax, variables, data types, conditionals, loops, execution flow, stack & heap intuition, reading tracebacks, debugging mindset	Calculator, number guessing game
Week 2	Functions & Data Structures	Functions, arguments, return values, scope, recursion basics, lists, dicts, tuples, sets, mutability, time complexity intuition	Student marks analyzer, to-do app
Week 3	Writing Real Python Code	Modules, imports, virtual environments, file I/O, exceptions, logging basics, debugging tools, writing scripts for automation	File organizer or CSV processor
Week 4	Git & Collaboration	Git internals, git add/commit/push/pull, branches, merge vs rebase, PRs, code reviews, README writing	Push Month-1 projects to GitHub

MONTH 2: Data, Text & Embeddings Foundations

Week	Focus	What You Learn (Detailed)	Build / Output
Week 5	Pandas & EDA	Pandas internals, CSV/Excel ingestion, indexing, filtering, groupby, joins, data leakage intuition, exploratory analysis	Dataset analysis notebook
Week 6	Data Cleaning	Missing value strategies, duplicates, outliers, feature engineering, normalization vs standardization, data pipelines	Cleaned dataset
Week 7	Text Processing	Text normalization, tokenization, stopwords, stemming vs lemmatization, regex basics, NLP preprocessing pipelines	Text cleaning pipeline
Week 8	Embeddings & Similarity	Embeddings theory, dense vs sparse vectors, cosine similarity, FAISS basics, OpenAI & HF embeddings APIs	Semantic search tool

MONTH 3: Machine Learning Foundations

Week	Focus	What You Learn (Detailed)	Build / Output
Week 9	ML Basics & Framing	ML vs rules, supervised/unsupervised learning, training/testing split, leakage, real-world ML use cases	ML problem definition
Week 10	Core ML Models	Linear & logistic regression, KNN, decision trees, bias intuition, model selection heuristics, scikit-learn pipelines	Regression/classification model
Week 11	Model Evaluation	Accuracy, precision, recall, F1, ROC-AUC, confusion matrix, cross-validation, error analysis	Evaluation report
Week 12	ML Best Practices	Overfitting/underfitting, bias-variance tradeoff, hyperparameter tuning, feature importance, end-to-end ML workflow	ML mini-project

MONTH 4: Generative AI & Prompt Engineering

Week	Focus	What You Learn (Detailed)	Build / Output
Week 13	LLM Fundamentals	Transformer basics, attention intuition, tokens, context windows, inference, latency vs quality tradeoffs	Basic chatbot
Week 14	Prompt Engineering	Role-based prompting, chain-of-thought (conceptual), few-shot prompting, structured outputs (JSON), prompt iteration	Summarizer
Week 15	GenAI APIs	OpenAI, Azure OpenAI, Hugging Face APIs, streaming responses, retries, rate limits, cost management	Resume reviewer
Week 16	Responsible AI	Hallucinations, bias, prompt injection, jailbreaks, output validation, guardrails, safety filters	Constrained chatbot

MONTH 5: RAG, Vector Stores & Agent Frameworks

Week	Focus	What You Learn (Detailed)	Build / Output
Week 17	RAG Fundamentals	Why RAG exists, RAG vs fine-tuning, chunking strategies, metadata filtering, retrieval failures	Chunking pipeline
Week 18	Vector Databases	FAISS, Chroma, Pinecone basics, ANN search, indexing strategies, similarity metrics	Vector search system
Week 19	RAG Systems	Retrieval + generation flow, context injection, reranking, RAG evaluation (precision@k, recall@k)	Document Q&A bot
Week 20	Frameworks & Agents	LangChain, Semantic Kernel, CrewAI, SmolAgents, agent vs workflow, memory, tools & function calling	Multi-step AI workflow

MONTH 6: Agentic Systems, Production & Capstone

Week	Focus	What You Learn (Detailed)	Build / Output
Week 21	Agentic Workflows	Planning, reasoning, tool orchestration, CrewAI roles, SmolAgents loops, failure handling	Agent-based automation
Week 22	Evaluation & Optimization	LLM evaluation (LLM-as-judge), cost tracking, latency optimization, caching, batching	Evaluation report
Week 23	Production Readiness	FastAPI, async inference, logging, monitoring, prompt versioning, secrets, deployments	Production AI service
Week 24	Capstone Project	System design, RAG/agents integration, architecture diagrams, README, demos, tradeoffs	Final capstone